

Heat Transfer

Questions with Answer Keys

JEE Main 2020 Chapterwise

Physics

Q1 JEE Main 2020 - 3 September (Morning)

A 750Hz, 20V (rms) source is connected to a resistance of 100Ω , an inductance of 0.1803H and a capacitance of $10\mu\text{F}$ all in series. The time in which the resistance (heat capacity $2\text{J}/^\circ\text{C}$) will get heated by 10°C . (assume no loss of heat to the surroundings) is close to

- (A) 348 s
- (B) 418s
- (C) 245s
- (D) 365s

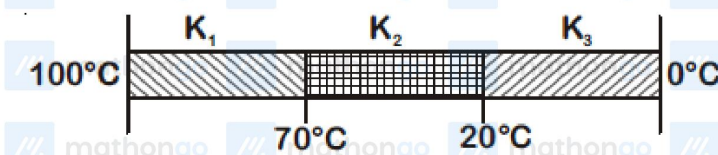
Q2 JEE Main 2020 - 3 September (Evening)

A metallic sphere cools from 50°C to 40°C in 300s. If atmospheric temperature around is 20°C , then the sphere's temperature after the next 5 minutes will be close to

- (A) 28°C
- (B) 35°C
- (C) 33°C
- (D) 31°C

Q3 JEE Main 2020 - 6 September (Evening)

Three rods of identical cross-section and lengths are made of three different materials of thermal conductivity K_1 , K_2 and K_3 , respectively. They are joined together at their ends to make a long rod (see figure). One end of the long rod is maintained at 100°C and the other at 0°C (see figure). If the joints of the rod are at 70°C and 20°C in steady state and there is no loss of energy from the surface of the rod, the correct relationship between K_1 , K_2 and K_3 is



- (A) $K_1 : K_3 = 2 : 3$
- (B) $K_1 < K_2 < K_3$
- $K_2 : K_3 = 2 : 5$
- (C) $K_1 : K_2 = 5 : 2$

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(D) $K_1 > K_2 > K_3$

$K_1 : K_3 = 3 : 5$

Q4 JEE Main 2020 - 7 January (Evening)

M grams of steam at 100°C is mixed with 200g of ice at its melting point in a thermally insulated container. If it produces liquid water at 40°C [heat of vaporization of water is 540 cal/g and heat of fusion of ice is 80 cal/g], the value of M is

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Answer Key

Q1 (A)

Q2 (C)

Q3 (A)

Q4 (40)

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Hints & Solutions

JEE Main 2020 Chapterwise

Physics

Q1

$$\begin{aligned} H &= (I_{rms})^2 R t \\ m S \Delta T &= (I_{rms})^2 R t \\ &= \frac{m S \Delta T}{(I_{rms})^2 R} \\ &= \frac{2 \times 10}{(I_{rms})^2 \times 100} \\ I_{rms} &= \frac{20}{\sqrt{(X_L - X_C)^2 + R^2}}, \quad X_L = 2 \times \pi \times 750 \times .1803 \\ X_C &= \frac{1}{2 \times \pi \times 750 \times 10^{-6}} \\ \text{Put value in (i), } t \text{ come out } &\simeq 348 \text{ s} \end{aligned}$$

Q2

$$\begin{aligned} \frac{\Delta T}{\Delta t} &= -b (T_{av} - T_s) \\ \therefore \frac{50-40}{5} &= -b [45 - 20] \\ \text{and, } \frac{40-T_3}{5} &= -b \left[\frac{40+T_3}{2} - 20 \right] \\ \Rightarrow T_3 &= \frac{200}{6} \approx 33^\circ \text{C} \end{aligned}$$

Q3

$$\begin{aligned} K_1(100 - 70) &= K_2(50) \\ K_3(20) &= K_2(50) \end{aligned}$$

Q4

$$\begin{aligned} M_{ice} L_f + m_{ice} (40 - 0) C_w &= m_{steam} L_v + m_{steam} (100 - 40) C_w \\ \Rightarrow 200 [80 + 40 (1)] &= m [540 + 60 (1)] \\ \Rightarrow 200 (120) &= m (600) \\ m &= 40 \text{ gm} \end{aligned}$$